

## Week 1 Big Data Analytics Lab Report

During Week 1 of the Introduction to Big Data Analytics course, I used Jupyter Notebook to practice basic Python programming and perform simple data analysis. I worked with **variables and data types, strings, lists, and dictionaries**. I also practiced using Python functions to calculate values such as **averages, maximums, and minimums**.

For the data analysis part, I loaded the provided Excel dataset containing information about 20 AUCA students into Python using Pandas. I converted the data into a list of dictionaries and analyzed students' scores and districts. **I calculated the class average, highest and lowest scores, score range, and the number of students from different districts.** I also identified the students with the highest and lowest scores.

One thing that surprised me was how easily Python could analyze a dataset using simple built-in functions such as **sum()**, **max()**, **min()**, and **len()**. This lab helped me understand how basic Python programming can be applied to real data and gave me a foundation for further learning in Big Data Analytics.